System Operations and Operation Contracts

Operation 1: request Appointment(patientID, professionalID, preferredTime, reason)

UC-01 Request / Book Appointment

Preconditions:

- The actor is authenticated and authorized to request an appointment.
- The patient can be identified in the system.
- The appointment request contains valid information.

Postconditions:

- The system retrieves available appointment options matching the request, if any exist.
- If no suitable appointments exist, the system returns a message and may suggest alternatives.
- No appointment is created or confirmed by this operation alone.

Operation 2: selectSlot(slotID)

UC-01 Request / Book Appointment

Preconditions:

- The actor has initiated an appointment request.
- The selected slot exists and was presented as an available option.
- The actor is authorized to book the appointment for the patient.

Postconditions:

- The system identifies the selected appointment slot.
- The system checks that the slot is still available.
- If the slot is unavailable, the system returns an error and requests a new slot to be selected.
- No appointment is created and confirmed until the booking is successfully completed.

Operation 3: confirmAppointment(patientId, professionalId, slotId, reason)

UC-01 Request / Book Appointment

Preconditions:

- The patient and medical professional exist in the system.
- The selected slot exists and is available.
- The slot belongs to the professional that was selected and is suitable for the requested appointment.
- The actor is authorized to create the appointment.
- Any required booking conditions are satisfied.

Postconditions:

- A new Appointment instance is created with a unique appointmentID.
- The appointment is associated with the patient and medical professional.
- The appointment's date and time correspond to the slot that was selected.
- The appointment status is confirmed.
- The successful booking is recorded in the system.

Operation 4: recordConsultation(appointmentId, observations, diagnosis, treatments, followUp)

UC-02 Record Consultation

Preconditions:

- The medical professional is authenticated and authorized to record clinical information.
- The appointment exists and is valid for consultation.
- The required consultation information is provided and valid.

Postconditions:

- A new Consultation instance is created with a unique consultationID.
- The consultation's date and time are recorded.
- The consultation is associated with the right appointment.
- The observations, diagnosis, treatments, and follow-up information are saved.
- The identity of the medical professional who recorded the consultation is saved for accountability.

Operation 5: createReferral(patientId, clinicalReason, priority)

UC-03 Create Referral

Preconditions:

- The patient exists in the system.
- The medical professional is authenticated and authorized to create referrals.
- The clinical reason and priority are valid.
- Any other mandatory referral information is provided.

Postconditions:

- A new Referral instance is created with a unique referralID.
- The referral is associated with the patient.
- The referral is associated with the medical professional.
- The clinical reason, priority, creation date, and referral status are recorded.
- The referral is saved and available to authorized users.